

Workforce Attrition Patterns and Risk Hotspot Analysis at Palo Alto Networks

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Abstract—Employee attrition is an important workforce management concern that can affect organizational stability, workforce planning, knowledge retention, and operational continuity. This study presents a diagnostic analysis of employee attrition using a dataset containing 1,470 employee records and 31 variables. Attrition patterns are examined across departments, job roles, demographics, tenure, overtime, business travel, and distance from home. The analysis identifies workforce segments with comparatively higher observed attrition rates rather than predicting individual employee departures. Key findings include an overall attrition rate of 16.12%, higher attrition among early-tenure employees, employees working overtime, frequent business travellers, younger employees, and selected job roles. An interactive Streamlit dashboard was developed to enable dynamic exploration of these patterns using workforce filters. The findings provide a data-driven basis for identifying attrition hotspots and areas requiring further workforce investigation.

Index Terms—employee attrition, workforce analytics, exploratory data analysis, employee retention, data visualization

I. INTRODUCTION

Employee attrition can affect workforce stability, operational continuity, knowledge retention, and workforce planning. Understanding attrition therefore requires examining not only the overall number of employee exits but also the workforce segments in which exits are concentrated.

Workforce analytics provides a data-driven approach for identifying such patterns. Previous research has examined employee turnover through job satisfaction, organizational commitment, organizational support, human resource practices, and workplace conditions [5], [7], [11], [14].

This study analyzes workforce attrition across organizational, demographic, tenure, workload, travel, and commuting dimensions. The objective is diagnostic rather than predictive: the study identifies segments with comparatively higher observed attrition rates rather than predicting whether an individual employee will leave.

An interactive Streamlit dashboard was also developed to allow users to explore these patterns dynamically through department, job-role, tenure, overtime, and business-travel filters.

II. PROBLEM STATEMENT AND OBJECTIVES

Attrition is not necessarily distributed uniformly across an organization. Different departments, job roles, employee groups, tenure levels, and working conditions may exhibit different observed attrition rates. Identifying these differences

can help organizations determine areas requiring further investigation.

The objectives of this study are to:

- 1) Measure overall employee attrition and retained versus exited employees.
- 2) Identify departments and job roles with higher observed attrition.
- 3) Examine attrition across age, gender, and education groups.
- 4) Analyze attrition across employee tenure groups.
- 5) Examine differences associated with overtime and business travel.
- 6) Analyze the relationship between distance from home and observed attrition.
- 7) Develop an interactive dashboard for workforce attrition analysis.

III. LITERATURE REVIEW

Employee turnover has been studied extensively in organizational and human resource research. Early models proposed that turnover results from a combination of organizational conditions, employee perceptions, job satisfaction, and perceived alternatives [1]–[3]. Later research expanded these models through the study of turnover intention, organizational commitment, and employee withdrawal processes [5], [6], [9].

Organizational support and human resource practices have also been associated with employee retention and turnover outcomes [7], [11], [12]. Research has further examined turnover at group and organizational levels, demonstrating that employee exits can have broader organizational consequences [?], [15], [17].

Recent literature has increasingly incorporated systematic reviews and data-driven methods into turnover research. Bolt et al. reviewed 1,375 turnover studies and highlighted the evolution of turnover research across theoretical and methodological approaches [19]. Machine-learning research has also investigated employee turnover prediction using employee-level data [20], [21].

Unlike predictive studies, the present work focuses on diagnostic exploratory analysis of observed attrition. It examines multiple workforce dimensions simultaneously to identify attrition hotspots and presents the results through an interactive dashboard.

IV. DATASET AND METHODOLOGY

The dataset contains 1,470 employee records and 31 variables covering demographic, organizational, employment, workload, travel, and commuting characteristics. The target variable, *Attrition*, is binary, where 0 represents retained employees and 1 represents employees who exited.

The dataset was checked for missing values and duplicate records. No missing values or duplicate records were identified. Additional categorical groups were created for age, tenure, and distance from home to support comparative analysis.

The attrition rate for each workforce segment was calculated as:

$$AR = \frac{N_{\text{exited}}}{N_{\text{employees}}} \times 100 \quad (1)$$

Exploratory Data Analysis was performed across department, job role, age, gender, education field, tenure, overtime, business travel, and distance from home.

An interactive dashboard was developed using Python, Streamlit, Pandas, and Plotly. Filters for department, job role, tenure, overtime, and business travel dynamically update the displayed workforce metrics and visualizations.

V. RESULTS AND DISCUSSION

The overall attrition rate in the dataset was 16.12%, with 237 employees exiting and 1,233 employees retained.

A. Organizational Patterns

Attrition varied across departments. Sales recorded the highest departmental attrition at 20.63%, followed by Human Resources at 19.05% and R&D at 13.84%.

At the job-role level, Sales Representatives recorded an observed attrition rate of 39.76%, identifying the role as a notable attrition hotspot.

The dashboard also allows users to combine department and job-role filters to examine more specific workforce segments.

B. Demographic Patterns

Age, gender, and education field were examined to understand demographic differences in observed attrition.

The 18–25 age group recorded the highest observed attrition rate at 35.77%. The gender analysis showed attrition rates of 14.80% for female employees and 17.01% for male employees.

The education-field analysis was incorporated into the dashboard to enable comparison across different educational backgrounds.

C. Tenure Patterns

Employee tenure showed a clear difference in observed attrition across groups.

Employees with 0–2 years at the company recorded an attrition rate of 29.82%. The corresponding rates for employees with 3–5, 6–10, and 11+ years of tenure were 13.82%, 12.28%, and 8.13%, respectively.

This indicates that early-tenure employees represented a comparatively higher attrition segment within the dataset.

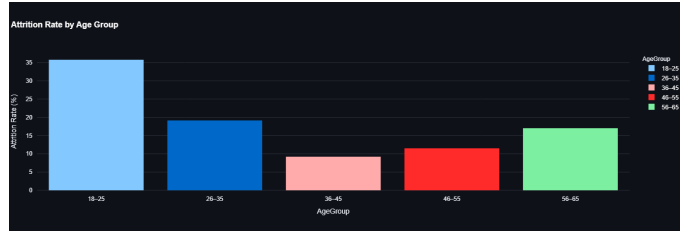


Fig. 1. Attrition rate by age group.

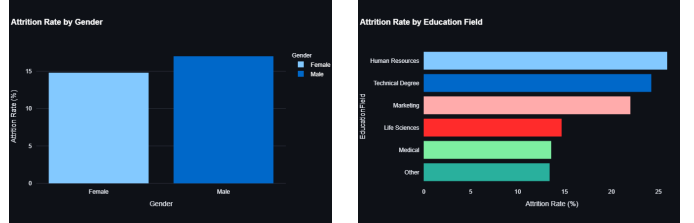


Fig. 2. Attrition rates by gender and education field.

D. Workload and Overtime

Overtime was associated with a substantial difference in observed attrition.

Employees working overtime recorded an attrition rate of 30.53%, compared with 10.44% among employees who did not work overtime.

This pattern highlights overtime as an important workforce dimension for further investigation, particularly in teams where extended working hours may be persistent.

E. Business Travel

Business travel also showed differences in observed attrition.

Employees classified as frequent business travellers recorded an attrition rate of 24.91%, compared with 14.91% for employees who travelled rarely and 8.00% for non-travelling employees.

These differences indicate that travel frequency may represent an additional workforce dimension that warrants further investigation in relation to workload and work-life balance.

F. Distance from Home

Distance from home was analyzed by grouping employees into four commuting-distance categories.

Observed attrition rates were 13.77%, 14.47%, 20.00%, and 22.06% for the 0–5, 6–10, 11–20, and 21+ distance groups, respectively.

The analysis shows an increasing observed attrition rate across the larger distance categories. However, this relationship is descriptive and does not establish that commuting distance directly causes employee attrition.

G. Overall Discussion

Overall, the analysis indicates that attrition was concentrated differently across organizational and employee segments.

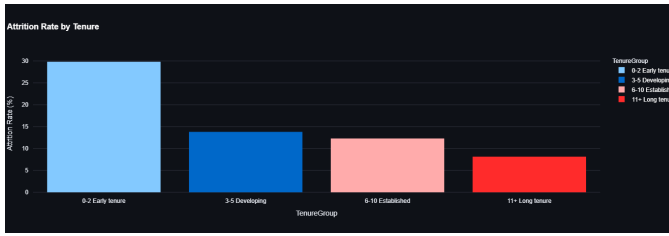


Fig. 3. Attrition rate by employee tenure.

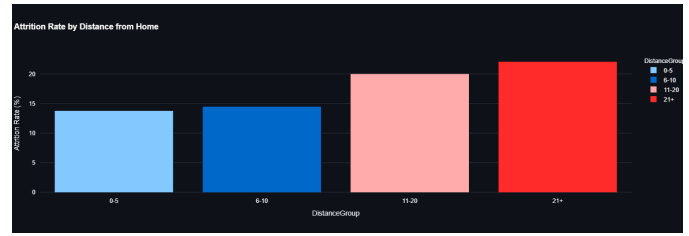


Fig. 5. Attrition rate by distance from home.

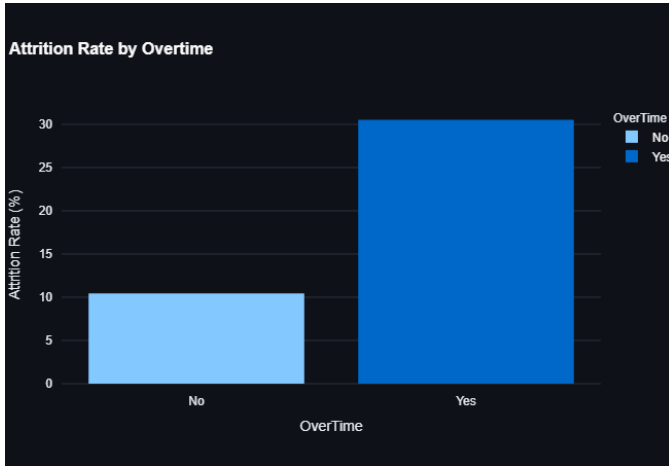


Fig. 4. Attrition rate by overtime status.

Higher observed attrition was found among early-tenure employees, employees working overtime, frequent business travellers, younger employees, employees living farther from the workplace, and selected job roles.

These relationships are descriptive and should not be interpreted as causal relationships. The dashboard enables users to investigate these patterns further by applying multiple filters simultaneously.

VI. RECOMMENDATIONS

The identified attrition hotspots suggest several areas for further workforce investigation.

- **Early-tenure retention:** Review onboarding, training, role clarity, mentoring, and employee support during the first two years.
- **High-attrition roles:** Examine workload, career progression, performance expectations, and working conditions in roles such as Sales Representative.
- **Overtime:** Investigate teams with consistently high overtime requirements and assess workload distribution.
- **Business travel:** Examine travel frequency and its relationship with workload and work-life balance.
- **Commuting:** Investigate whether employees living farther from the workplace face additional retention challenges.
- **Continuous monitoring:** Use the interactive dashboard to monitor changes in attrition across departments and workforce segments.

These recommendations represent areas for further investigation rather than causal conclusions derived from the analysis.

VII. LIMITATIONS

This study has several limitations.

First, the analysis is descriptive and does not establish causal relationships between employee characteristics and attrition.

Second, the findings are limited to the available dataset and may not generalize to other organizations or time periods.

Third, variables such as employee satisfaction, compensation, management quality, and organizational culture are not sufficiently represented in the analysis.

Finally, the study does not generate individual-level attrition predictions.

Future work could incorporate additional workforce variables, longitudinal data, statistical testing, or predictive modeling to further investigate the observed relationships.

VIII. CONCLUSION

This study analyzed workforce attrition patterns across organizational, demographic, tenure, workload, travel, and commuting dimensions using an employee-level dataset of 1,470 records.

The overall attrition rate was 16.12%, while several workforce segments exhibited substantially higher observed rates.

Notable patterns included higher attrition among Sales Representatives, younger employees, early-tenure employees, employees working overtime, frequent business travellers, and employees living farther from the workplace.

An interactive Streamlit dashboard was developed to support dynamic exploration of these patterns through department, job-role, tenure, overtime, and business-travel filters.

The study demonstrates that examining attrition across multiple workforce dimensions can provide more detailed insight than relying on an organization-wide attrition rate alone. The identified patterns provide areas for further investigation and support data-informed workforce retention planning.

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